

Documents to a reviewable entry.

SWE intern take-home | TypeScript frontend + backend | 6 October 2026

01 / FIND THE CONTRACT BEFORE GENERATING XML

I traced NetCHB's public service directory to the EntryUploadService WSDL and the real entry XSD, then vendored both schemas for offline validation. uploadEntry takes username, password and an entryXml string over SOAP 1.1 RPC/literal. The app creates the inner entry document in NetCHB's namespace, respects its root order, uses system-generated entry numbering and validates with xmllint. Upload acceptance is distinct from CBP acceptance; the app does not transmit entries.

02 / EXTRACT FACTS, KEEP THE EVIDENCE

PDF.js reads embedded text. Poppler rasterizes scanned pages; deskewing and table-rule removal improve Tesseract OCR. A conservative local parser works without credentials. An optional AI adapter handles varied layouts using page text and a structured contract. Zod validates output; unsupported page quotes are discarded. Documents are treated as untrusted data, not instructions. Selected values remain editable while original evidence is retained.

03 / RECONCILE THE SHIPMENT, SURFACE THE GAPS

The supplied invoice has five lines across two pages. The printed commercial amounts total \$22,875, versus \$23,235 stated FOB. The \$360 difference matches the T-shirt price mismatch: $2,400 \times \$2.85 = \$6,840$, versus \$6,480 printed. Hoodies disagree on quantity (1,200 versus 1,176) and blend (60/40 versus TC 65/35). Packing weight is 1,888 kg versus 1,930 kg on the B/L. These remain explicit review items, not silent corrections.

Provisional customs value: \$27,079

\$22,875 printed merchandise + \$4,180 buyer-fabric assist + \$24 FOC samples.

If the T-shirt amount is corrected to \$6,840, the value becomes \$27,439.

Freight \$3,850 and insurance \$185 are separate; their exclusion needs actual-cost evidence.

Bangladesh samples retain their specific origin and factory, despite blanket Vietnam declarations. Visual review corrected an OCR-transposed master bill and proposed the handwritten container replacement; these corrections are stored separately from automatic extraction. Tax IDs, ports, actual dates, bond/payment settings, ten-digit HTS codes, MIDs and statistical units are not invented. The sample XML is intentionally incomplete until broker input and discrepancy resolution.

04 / VERIFY, THEN IMPROVE

Ten regression tests cover the sample, an unseen EUR/Italy invoice, evidence rejection, review invalidation, missing-field blockers, escaping and actual XSD pass/fail. The supplied PDFs were tested through the upload API and browser. The live AI provider and hosted deployment were not tested. With more time: build a labeled unseen-document evaluation set; add vision-assisted handwriting and field-level confidence; isolate OCR workers with timeouts; implement current HTS/remedy/PGA rules and more entry types; add persistent audit history, authentication and an authorized NetCHB sandbox adapter.

Primary references: netchb.com/xml/entry/entry.xsd; NetCHB EntryUploadService WSDL; CBP rulings H354576 and 546363. AI use: Codex generated and iterated the implementation; tests and source inspection checked its work. Complete visible chat/tool history accompanies the submission.